Project Review 2 · MILESTONE 2

(Project Review 2)
Data Understanding & Cleaning

Present your dataset overview, your cleaning decisions, and prove your data is now trustworthy.

WELCOME BACK

Quick Recap of Module 2

1. NumPy arrays and vectorized operations
2. Broadcasting, reshaping, and aggregation
3. Pandas Series, DataFrames, and CSV loading
4. Sorting, grouping, merging, and pivot tables
5. Detecting and handling missing values and duplicates
6. Fixing data types and handling outliers

TODAY'S GOAL

There is no new coding today. Present your dataset understanding and cleaning decisions, and get feedback before analysis begins.

Today's Agenda

1. Welcome & format
2. What Milestone 2 covers
3. What you must submit
4. How you're evaluated
5. Demo day flow
6. What's next

No new coding today — this session is your second project checkpoint.

Why This Milestone Matters

◎ **DEFINITION**
Every later milestone assumes your dataset is clean and well understood.

1. A dirty dataset quietly produces wrong statistics, charts, and models later on
2. Explaining your cleaning decisions out loud is what separates real understanding from guesswork
3. This is the last checkpoint before Module 3, where you'll visualize and prepare this same data

What Happens in This Project Review

◎ **DEFINITION**
Each team presents their dataset and cleaning work, and receives feedback on the spot.

1. Your team walks through your dataset overview and cleaning decisions
2. The instructor and your peers ask questions about specific choices you made
3. You are graded on understanding and justification, not on a perfectly polished report

PART 2 · WHAT YOU SUBMIT

Your Analysis of Data Quality

1. A missing values analysis — what's missing, and how much

2. A duplicate analysis — how many, and how you found them

PART 2 · WHAT YOU SUBMIT

Your Cleaning Work

1. Your data cleaning decisions, and the reasoning behind each one
2. Any remaining data quality issues you chose not to fix, and why
3. Summary statistics of your dataset before and after cleaning

PART 2 · WHAT YOU SUBMIT

Your Deliverables

1. A clean dataset, saved and pushed to GitHub
2. A cleaning notebook showing every step you took
3. Documentation explaining your decisions in plain language

Example: Documenting a Cleaning Decision

cleaning_notebook.py

```
# Decision: filled missing 'age' values with the median (34)
# Reason: only 12 of 800 rows were missing; dropping them
# would lose data with no clear pattern to why it's missing.
df["age"] = df["age"].fillna(df["age"].median())
```

A short comment like this, for every decision, is exactly what your cleaning notebook should contain.

Before-and-After Snapshot

Metric	Before → After
Rows	800 → 793
Missing values	16 → 0
Duplicate rows	7 → 0
Max income value	412,000 → 98,750 (capped)

What Feedback Focuses On

1. Is your dataset understanding thorough? Do you know your data ?

2. Are your cleaning decisions justified? Can you defend each one?

3. Is your documentation clear? Could someone else follow your work?

JUST IN CASE

Common Milestone 2 Mistakes

Mistake	What Happens
Cleaning without checking first	Rows get dropped or values get changed before anyone confirms there's a real problem
No before-and-after comparison	It's impossible to tell how much the cleaning actually changed the dataset
Treating every outlier as an error	Some extreme values are real and meaningful — deleting them can hide the truth
Undocumented decisions	The team can't explain their own cleaning steps when asked in the demo

QUICK RECAP

What a Strong Milestone 2 Looks Like

1. A clear dataset overview and data dictionary
2. Exact missing value and duplicate counts
3. Every cleaning decision explained with a reason
4. Outliers investigated, not deleted by default
5. A clean dataset and notebook pushed to GitHub
6. A team that can defend its choices under questions

Thank you